

A long time ago in a galaxy far, far away...

Y17 (2017) — English translation

Part A. Modification of Newtonian dynamics

The dependence of the orbital speed of stars and gas in a galaxy on the distance to the center of the galaxy is called the rotation curve of the galaxy. According to Newtonian dynamics, the speed should decrease with distance from the center of the galaxy (Fig. 1A), assuming that most of the galaxy's mass is located at its center. However, it turns out that in fact the speed is an almost constant function of the radius (Fig. 1B). One solution to this discrepancy can be obtained by modifying Newtonian dynamics as follows.

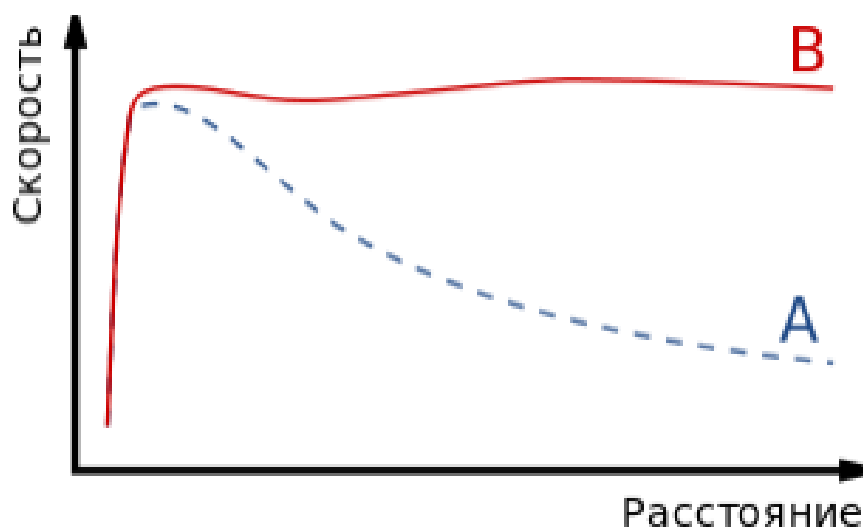


Figure 1: Figure 1.

Acceleration due to gravity depends not just on the mass m , but on $m \cdot \mu\left(\frac{a}{a_0}\right)$, where $\mu(x)$ is a certain function whose value tends to unity for large arguments ($x \gg 1$) and to x for small arguments ($x \ll 1$), a is the acceleration due to gravity, and a_0 is a constant. Thus, Newton's second law takes the form: $F = m \cdot \mu\left(\frac{a}{a_0}\right) a$. The centripetal acceleration of stars and gas clouds on the outskirts of spiral galaxies is, as observed, much less than a_0 . A graph of the actually measured rotation curve of the Milky Way is shown in Figure 2. Here 1 kpc = 1 kiloparsec and 1 parsec = $3 \cdot 10^{16}$ m.

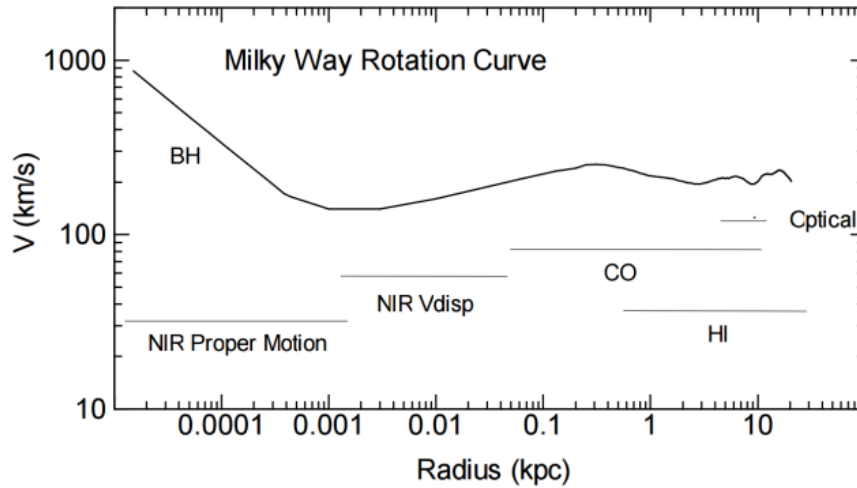


Figure 2: Figure 2.

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| A1 | How does the orbital rotation speed depend on the distance to the galactic nucleus in the case of ordinary Newtonian dynamics? Consider that the mass of the galaxy is concentrated in its core. | 0.50 |
| A2 | Let in reality the orbital rotation speed be constant v_∞ at large distances from the galactic center (as shown by the measured curve in Fig. 1B). Find the centripetal acceleration $a(r)$ of a star at distance r from the galactic center. Estimate v_∞ from the graph and compute a at $r = 10$ kpc. Compare this with the constant a_0 and discuss whether the MOND regime ($a \ll a_0$) is applicable at these distances. | 1.00 |
| A3 | Using the portion of the curve in the region $r < 1$ parsec, estimate the mass of the Milky Way core, assuming that Newtonian dynamics operates near the core at distances $r < 1$ parsec from the galactic center. Assume that the main contribution to gravity is made only by the mass in the core. | 1.00 |
| A4 | Taking into account the modified Newtonian dynamics described above, prove that at large distances from the galactic nucleus, where the acceleration due to gravity is small, the orbital speed does not depend on the distance to the center and is equal to a constant. Let this speed far from the center be equal to v_∞ . How are the constants a_0 and v_∞ related? Using the graph, estimate v_∞ . | 1.00 |
| A5 | Estimate the value of the constant a_0 based on the data in the graph, assuming that in reality the mass of the entire Milky Way is $k = 10^6$ times greater than the mass of the core. | 0.50 |

Part B. Tidal forces

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| B1 | It is known that the power of tidal frictional forces acting on the Earth, caused by the Moon's revolution around it, is $N \approx 10^{12}$ W. Estimate, by order of magnitude, the increment in the duration of the Earth's day ΔT_E and the lunar month ΔT_l over a time $\tau = 100$ million years. The average radius of the Moon's orbit is $R = 4 \cdot 10^5$ km, the mass of the Moon is $m_M = 10^{22}$ kg, the orbital period of the Moon is $T_M = 27.3$ days, and the moment of inertia of the Earth is $I = 8 \cdot 10^{37}$ kg m ² . The orbital motion of the Moon is co-directed with the rotation of the Earth around its axis. | 3.00 |
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Part C. Relativistic processes

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| C1 | The relativistic π^0 -meson decayed into two gamma quanta with energies $E_1 = 81$ MeV and $E_2 = 450$ MeV. Find the angle φ between the directions of the two gamma quanta. The rest mass of the π^0 -meson is $135 \text{ MeV}/c^2$. | 1.00 |
| C2 | A positron with Lorentz factor $\gamma = 3$ collides with an electron at rest. The resulting two gamma quanta scatter symmetrically relative to the direction of the positron. Find the angle φ between the directions of the two gamma quanta. | 2.00 |

Source: <https://pho.rs/p/3041>